

Linux Commands

Networking Commands

Network Basics Commands:

Command	Description	Variants	Example	Output / Use Case
ping	Check if a host is reachable and measure latency.	-c (count), -i (interval)	ping -c 4 google.com	Connection test, packet loss check.
ifconfig	Shows network interfaces & IP addresses. (Old tool)	up, down	ifconfig eth0	Debugging network interface issues.
ip addr	Modern command to show IP, routes, link state.	ip a, ip link, ip route	ip addr show	Check IP configuration, troubleshoot networking.
netstat	Shows network connections & listening ports.	-tulpn	netstat -tulpn	List active services & ports.
ss	Modern replacement for netstat (faster).	-tulpn, -s	ss -tulpn	Identify which process is listening on which port.
curl	Fetch URLs, APIs, website data.	-I headers, -o output	curl -I http://example.com	DevOps me API testing & health checks.
wget	Download files from internet.	-c continue, -O output file	wget https://file.com/app.deb	Download packages/programs.
tracert	Shows route packets take to a destination.	-n (no DNS)	tracert google.com	Diagnose slow network paths.
nslookup	Query DNS records.	(none)	nslookup youtube.com	DNS troubleshooting.

Firewall & Ports Commands:

Command	Description	Variants	Example	Output / Use Case
ufw	Simple firewall tool on Ubuntu.	enable, allow, deny	ufw allow 8080	Allow/deny ports for servers.
iptables	Advanced firewall tool.	-A, -D, -L	iptables -L	Deep control over packet filtering.
firewalld	Dynamic firewall used in RHEL/CentOS.	--add-port, --zone	firewall-cmd --add-port=80/tcp	Open ports without restarting firewall.
nmap	Scan open ports on a host.	-sV version scan	nmap -sV localhost	Security audit, port discovery.

SSH & Remote Access Commands:

Command	Description	Variants	Example	Output / Use Case
ssh	Connect securely to remote server.	-i key file	ssh user@192.168.1.10	Remote login, DevOps main tool.
scp	Secure copy files between machines.	-r recursive	scp file.txt user@server:/home/	Transfer backups, configs.
rsync	Sync files faster than SCP.	-avz	rsync -avz folder/user@server:/srv/	Deployments, large file sync.

Hostname & DNS Commands:

Command	Description	Variants	Example	Output / Use Case
/etc/hosts	Local manual name-to-IP mapping file.	—	Update file	Map myserver.local → 127.0.0.1
hostnamectl	Change/view hostname.	set-hostname	hostnamectl set-hostname dev-machine	Permanent hostname changes.